

Lesson 12. Introduction to Calculus of variations

ELTE Eötvös Loránd University

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Introduction

Classical Analysis: One Variable Function

- Given: $f(x) : \mathbb{R} \rightarrow \mathbb{R}$
- Find x for min/max
- Necessary condition:
 - Solve: $f'(x) = 0$
 - Solution x^* - stationary points are candidates for local min, max, or inflection point
- Sufficient condition:
 - $f'' > 0$: local min
 - $f'' < 0$: local max
 - $f'' = 0$: inconclusive

Classical Analysis: Multi-variable Function

- Given: $f(\mathbf{x}) : \mathbb{R}^2 \rightarrow \mathbb{R}$
- Find $\mathbf{x}$ for min/max
- Necessary condition:
 - Solve: $\nabla f = 0$
 - Solution $\mathbf{x}^*$ - stationary points are candidates for local min, max, or saddle point
- Sufficient condition:
 - H pos. def.: local min
 - H neg. def.: local max
 - H indef.: saddle

Calculus of Variations

- Given:

$$J[y] = \int_a^b F(x, y, y') dx$$

- Find $y(x)$ for min/max
- Necessary condition:
 - Solve: $\delta J = 0$
 - Euler-Lagrange:

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

- Solution $Y(x)$ - an extremal of the functional.
- Sufficient condition:
 - Classify via $\delta^2 J$

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Functionals

A **functional** is a real-valued function:

$$J[y] : S \rightarrow \mathbb{R}$$

whose domain S is a set of functions.

What is a Functional

In other words, a **functional** is a special kind of mathematical object. It's not a function from real numbers to real numbers like $f(x) = x^2$. Instead, a functional takes a **function as input** and **returns a number**.

We say that $J(y)$ has an **extremum** when $J(y)$ takes its maximum or minimum value.

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Functionals

Example:

Let $\mathcal{C}[0, 1]$ be the set of all continuous functions defined on the interval $[0, 1]$. Define the functional

$$J[y] = \int_0^1 y(x) dx.$$

This means: give a function $y(x)$ to functional, and it returns a number - the **area under the curve** of $y(x)$ from $x = 0$ to $x = 1$.

Functionals

Consider

Function $y(x)$	Value of $J[y] = \int_0^1 y(x) dx$
x	0.5
x^2	0.333
x^3	0.25
$\sin x$	0.4597
$\cos x$	0.8415
e^x	1.7183
1	1

Functionals

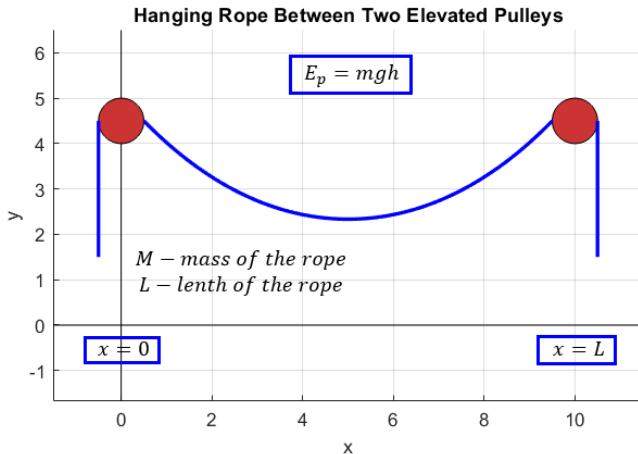
Conclusion:

We can now ask: *for which function $y(x)$ does the functional $J[y]$ attain its maximum or minimum value?* Among the seven functions given in the table, the functional $J[y] = \int_0^1 y(x) dx$ attains its minimum value when $y(x) = x^3$, and its maximum value when $y(x) = e^x$.

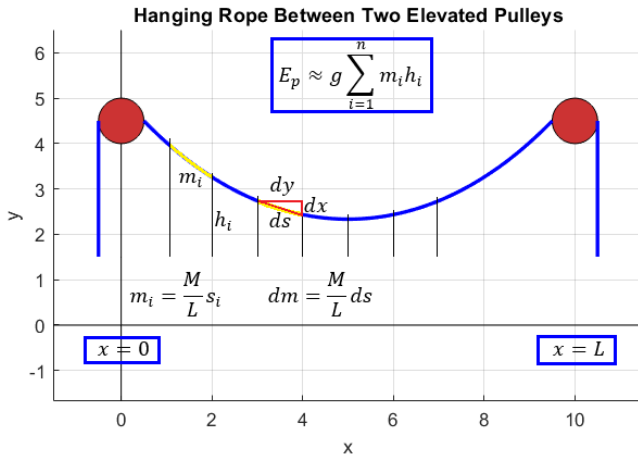
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Example 1: Minimizing the Gravitational Potential Energy of a Hanging Rope



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The total gravitational potential energy becomes a sum

Let's divide the rope into many small pieces. Then the total gravitational potential energy of the rope becomes the sum of the potential energy of each segment:

$$E_p \approx g \sum_{i=1}^n m_i h_i$$

Here:

- m_i is the mass of the i -th segment,
- h_i is the vertical position (height) of the center of that segment,
- g is the gravitational acceleration.

Example 1: Minimizing the Gravitational Potential Energy of a Hanging Rope

The length of a small segment

Assuming the rope has total mass M and total length L , and the mass is uniformly distributed, the mass of each segment is proportional to its length. Thus:

$$m_i = \frac{M}{L} s_i$$

where s_i is the length of the i -th segment.

Example 1: Minimizing the Gravitational Potential Energy of a Hanging Rope

Length of a segment

The i -th segment stretches between two points on the graph of $y(x)$, say from x to $x + \Delta x$, and we can approximate its length using the Pythagorean theorem (as a straight-line approximation):

$$s_i \approx \sqrt{(\Delta x)^2 + (\Delta y)^2} = \Delta x \cdot \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2}$$

Example 1: Minimizing the Gravitational Potential Energy of a Hanging Rope

As $n \rightarrow \infty$, the changes in x and y become infinitesimal, and the length element becomes

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \sqrt{1 + (y')^2} dx$$

And the infinitesimally small change in mass becomes:

$$dm = \frac{M}{L} ds = \frac{M}{L} \sqrt{1 + (y')^2} dx$$

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The gravitational potential energy of this small mass element located at height $y(x)$ is:

$$dE_p = gy(x) dm = gy(x) \cdot \frac{M}{L} \sqrt{1 + (y')^2} dx$$

Total Gravitational Potential Energy

Then, using integral calculus, the total gravitational potential energy of the rope is given by the limit of the Riemann sum $g \sum_{i=1}^n m_i h_i$, or in other words, the integral from $x = 0$ to $x = L$:

$$E_p[y] = \frac{M}{L} \cdot \int_0^L gy(x) \cdot \sqrt{1 + (y'(x))^2} dx$$

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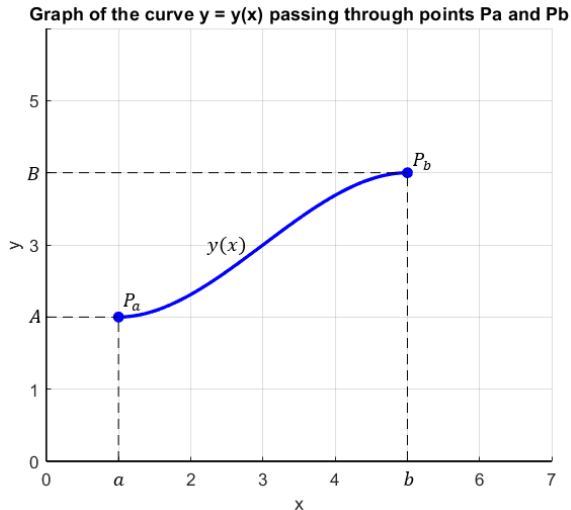
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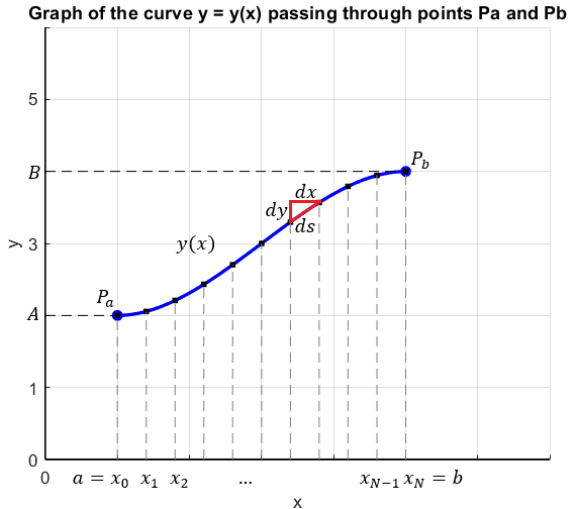
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Example 2: Shortest Distance Between Two Points



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We divide the interval $[a, b]$ into N subintervals of equal length:

$$\Delta x = \frac{b - a}{N},$$

with nodes $x_0 = a, \dots, x_N = b$ and $x_{k+1} - x_k = \Delta x$.

The curve is approximated by straight-line segments. The length of the k -th segment is:

$$\Delta s_k \approx \sqrt{(\Delta x)^2 + (\Delta y)^2} = \Delta x \cdot \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2}.$$

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Example 2: Shortest Distance Between Two Points

As $N \rightarrow \infty$, this becomes:

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \sqrt{1 + (y')^2} dx.$$

which tends to the integral:

$$S[y] = \int_a^b \sqrt{1 + (y')^2} dx.$$

$S[y]$ is a functional

This expression gives the length of the curve $y = y(x)$. The quantity $S[y]$ is a **functional** - it maps functions to numbers. This functional is minimized by the straight line joining P_a and P_b .

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General Form of Variational Problems

Find the function $y(x)$ that **extremizes the functional**:

$$J[y] = \int_{x_1}^{x_2} F(x, y, y') dx,$$

subject to the boundary conditions:

$$y(x_1) = y_1, \quad y(x_2) = y_2,$$

where F is a twice continuously differentiable function.

The **goal** is to find a function $y(x)$ for which $J[y]$ attains a minimum or maximum - the so-called **extremal function**.

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Variation of a Function

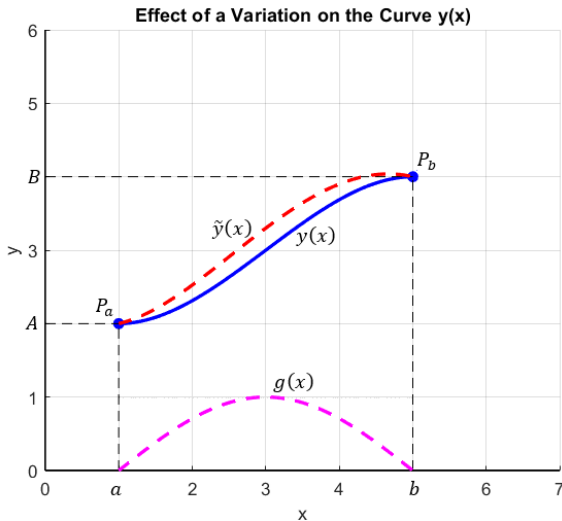
Definition (Variation of a Function): Given a function $y(x)$ that is believed to minimize some quantity (such as length, area, or energy), consider a perturbed version of this function:

$$y_{\varepsilon}(x) = y(x) + \varepsilon \cdot g(x)$$

where:

- ε is a small real parameter;
- $g(x)$ is a smooth function such that $g(x_1) = g(x_2) = 0$, so the perturbation does not affect the boundary values;
- $y_{\varepsilon}(x)$ is called a **variation** of the function $y(x)$.

Variation of a Function



First variation

Intuition

The **first variation** of a functional $J[y]$ is a quantity that tells us how $J[y]$ changes when we slightly perturb the function $y(x)$.

Given a perturbed function

$$y_\varepsilon(x) = y(x) + \varepsilon g(x),$$

with boundary conditions

$$g(a) = g(b) = 0,$$

the **first variation** is:

$$\delta J[y; g] = \left. \frac{d}{d\varepsilon} J[y + \varepsilon g] \right|_{\varepsilon=0}.$$

This is called the *Gateaux derivative* of the functional and is a linear functional of the perturbation $g(x)$.

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Fundamental Lemma of Calculus of Variations

Stationary functions

If

$$\delta J[y; g] = 0$$

for all such $g(x)$, then $y(x)$ is said to be a *stationary function* - a candidate for a minimum or maximum of the functional.

Fundamental Lemma of Calculus of Variations: Let $f(x)$ be continuous on $[a, b]$. If

$$\int_a^b f(x)g(x) dx = 0$$

for every smooth function $g(x)$ with $g(a) = g(b) = 0$, then:

$$f(x) \equiv 0 \quad \text{on } [a, b]$$

This means: if an integral vanishes for *all* functions $g(x)$ that are zero at the end-points, then the integrand itself must be identically zero.

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Euler-Lagrange Equation

Theorem:

If $y(x)$ is an extremizing function for the problem

$$\text{Minimize/Maximize } I(y) = \int_{x_1}^{x_2} F(x, y, y') dx,$$

with end conditions $y(x_1) = y_1$ and $y(x_2) = y_2$, then $y(x)$ satisfies the boundary value problem

$$\frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) - \frac{\partial F}{\partial y} = 0, \quad (1)$$

$$y(x_1) = y_1 \quad \text{and} \quad y(x_2) = y_2.$$

Equation (1) is known as the **Euler-Lagrange equation**.

Example 1

Find the extremals of the functional

$$\int_{x_0}^{x_1} \frac{y'^2}{x^3} dx$$

Solution

$$F(x, y, y') = \frac{y'^2}{x^3}; \quad \frac{\partial F}{\partial y} = 0; \quad \frac{\partial F}{\partial y'} = \frac{2y'}{x^3}$$

Euler-Lagrange equation

$$\frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = \frac{d}{dx} \left(\frac{2y'}{x^3} \right) = \frac{2x^3 y'' - 6y' x^2}{x^6} = \frac{2}{x^4} (xy'' - 3y') = 0$$

Thus we have

$$xy'' - 3y' = 0 \quad \text{or} \quad \frac{y''}{y'} = \frac{3}{x}$$

Example 1

Solution (cont.)

Integrating

$$\int \frac{y''}{y'} dx = 3 \int \frac{1}{x} dx + c$$

$$\ln y' = 3 \ln x + c$$

$$\ln \frac{y'}{x^3} = c \quad \text{or} \quad y' = Cx^3$$

$$y = \frac{Cx^4}{4} + C_2$$

$$y = Ax^4 + B$$

Example 2

Extremize the functional

$$I(y) = \int_{x_1}^{x_2} 1 + y'^2 dx, \quad y(x_1) = y(x_2) = 0$$

Solution

$$F = 1 + y'^2; \quad \frac{\partial F}{\partial y'} = 2y'$$

Euler Equation

$$\begin{aligned} \frac{d}{dx}(2y') &= 0 \implies 2y'' = 0 \\ y'(x) &= C, \quad y(x) = Cx + D \\ C &= \frac{y_1 - y_2}{x_1 - x_2}; \end{aligned}$$

Example 3

Find the curve y on which the functional

$$\int_0^1 y'^2 + 12xy \, dx$$

is extremum, with $y(0) = 0$, $y(1) = 1$.

Solution

Here $F = y'^2 + 12xy$

Euler Lagrange Equation: $12x - 2y'' = 0$ or $y'' = 6x$

$$y' = 3x^2 + C$$

$$y = x^3 + cx + c'$$

Applying the conditions we get

$$y = x^3$$